SURVEY

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Research Article

Enhancing Fog Offloading and Task Management in IoT-Fog-Cloud Environments: A Review of Cutting- Edge Algorithms, Networks, and SDN Applications

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ABSTRACT

Because Internet of Things (IoT) devices and other IT forms are used in almost every area of human life, there is a vast volume of data that has to be handled and saved. The only practical solution to this issue is to handle and store enormous volumes of data utilizing cloud computing platforms. Real-time data analysis has always been crucial. However, it becomes even more crucial as IoT and technology develop and new applications—autonomous cars, smart cities, and IoT gadgets for the medical industry, agriculture, and other sectors—appear. Because of the massive amount of data involved, moving to a distant cloud takes time and severely congests the network, making cloud management and fast data processing difficult. Near-device processing at the network edge is made feasible by fog computing, which also makes data analysis almost instantaneous. An important issue for fog nodes is the increasing quantity of IoT devices and the data they produce. Task offloading may enhance fog computing by transferring the processing of excess data to other nodes, since resources are scarce in the fog. Fog devices need the best possible resource and task management. This review article examines task offloading in an IoT-Fog-Cloud scenario. In addition, we discuss software-defined network (SDN) applications, fog networks, and fog offloading challenges.

I. INTRODUCTION

The amount of data produced by Internet of Things devices has increased exponentially as a result of the widespread development of intelligent devices and the growth of cloud computing.

Furthermore, for data processing and transmission, modern Internet of Things applications require extremely low latency. These consist of intelligent manufacturing, autonomous driving, virtual reality, and augmented reality. "Fog computing," an innovative computer framework, is intended to meet these requirements. Applications that are sensitive to latency, such as virtual reality (VR), augmented reality (AR), and online games, may be able to process vast volumes of data at fog nodes instead of sending it all to a centralized cloud server. Thus, the amount of data transferred inside the main networks and information processing latency would be greatly reduced. Fog nodes in fog computing are situated close to Internet of Things (IoT) devices in order to serve as a connection bridge between these devices and the public cloud. A fog node could be an IoT gateway, a cloudlet, or a micro-

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data center, depending on its architecture. Despite several practical and instructional initiatives, the full benefits of fog computing are still to be realized [1], [2]. The Internet of Things produces massive volumes of data, which can be managed and analyzed via cloud computing. Nonetheless, a lot of Internet of Things devices require real-time processing. Fog computing offers an outstanding solution to this issue.

The restricted resources in fogs for heavy-duty processing, especially for procedures that must be completed rapidly or in real-time, make offloading duties from one fog to another node essential for uninterrupted processing activities.

SDN and fog offloading have been applied extensively in a variety of sectors, demonstrating their ability to enhance operational effectiveness and decision-making. In order to reduce latency and power consumption, fog computing is used in smart cities. This is important for Internet of Things applications like public safety and traffic management.

Furthermore, SDN-enabled wireless fog networks provide efficient load balancing and lower latency [3, 4, 5]. In the industry, it enables faster data processing at the margins, improving the reliability and promptness of working. [6], [7]. Industrial automation benefits from fog computing and SDN because they offer predictive maintenance and process optimization, especially for industrial Internet of things (IoT) systems, offering excellent reliability and real-time performance.[8], [9].

SDN and fog computing have been integrated in vehicle networks [11]. and vehicular ad-hoc networks (VANET) [10], while fog computing, SDN, and blockchain are combined in [12], to improve IoT sensors in agriculture. Furthermore, in the field of Intelligent Transportation Systems (ITS), [13], emphasizes further creative implementations, while [14]. Concentrates on using distributed machine learning on a fog network to anticipate forest fires. These illustrations show how deeply technology has impacted a number of businesses. These illustrations show how strongly technology is impacting a number of different businesses. Several frameworks and techniques have been employed to offload computation from the fog onto other nodes in the computing infrastructure, according to an assessment of the available research.

Although algorithms provide several solutions to address this issue, such as offloading to the cloud, the most feasible method is to upload work to another fog because cloud offloading is quite expensive. The assignment at hand involves selecting the ideal node from among all others on the network: one with sufficient processing power, a fast enough connection, adequate bandwidth, and a reasonable distance from the originating location.

Both distributed and centralized architectures offer advantages and disadvantages, according to investigations on different approaches and techniques and an analysis of the available literature. Together with the other nodes and links in the network, the fogs put together their data. This endeavor must not only place a significant computational load on the fog and communication lines, but also increase the effectiveness of the computing fog and underlying network. Since the centralized design depends on a single device, there are performance problems. This is because a breakdown in communication between the nodes might cause outages.

This review article explores an in-depth investigation of the prevalent issues and their solutions in light of the many complications related to offloading in fog computing, with a focus on the complicated nature of network topologies. We want to explore the many facets of network use in fog offloading and investigate the uses, advantages, and challenges related to SDN technology in this field.

The main contributions of this survey are as follows:

1. A detailed examination of the fog offloading problem categorizes the latest fixes and determines the different factors influencing the offloading choice.
2. With a focus on SDN applications, we examine and evaluate various offloading network types.
3. To address the service placement problem for Internet of Things applications spread over fog nodes, we offer a categorization for optimization methodologies.
4. Lastly, we identify significant challenges and investigate potential future research areas in fog-based systems.

In the sections that follow, we will dive fully into each problem in this review article. We will look over the important elements, concepts, and aspects of fog computing in the second section. The linked surveys are covered in the third section. The fourth section will then include an in-depth analysis of current research and an assessment

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and evaluation of options linked to fog offloading.

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We shall analyze and rank the methods and classifications of SDN and fog computing architectures in the fifth section. We will talk about the findings in the sixth section, highlighting the benefits and issues related to SDN in fog computing. In the final section, we will draw findings that will open up new avenues for fog computing research.

II OVERVIEW OF FUNDAMENTAL CONCEPTS

A. IoT

The term "Internet of Things" (IoT) describes an extensive network of connected objects that are equipped with software, sensors, and other technologies to allow them to trade and gather data over the internet. These gadgets might be anything from everyday objects like wearable technology and home appliances to cars and industrial gear.

IoT's capability to link and interact with these devices is essential because it offers enormous opportunities for automation, data-driven decision-making, and efficiency improvements in a number of industries. Through real-time data collection from many sources, Internet of Things (IoT) devices may yield useful insights that stimulate innovation and enhance operating procedures. IoT ("Internet of Things") operates on several fundamental concepts:

Sensors and Connectivity: To gather information about temperature, motion, and location, devices in an Internet of Things network are outfitted with sensors. To send this data, they make use of a variety of connectivity options including Bluetooth, Wi-Fi, and cellular networks.

Data processing: The device's local processor or the cloud are used to process the collected data. To extract significant patterns and insights from the data, advanced analytics and machine learning techniques may be used to examine the data.

Control and Automation: IoT makes it possible to automate procedures and activities in response to data that is received. For instance, occupancy patterns picked up by sensors are used by smart thermostats to change the temperature.

Security and privacy: Strong security measures (such as authentication and encryption) are essential to guard against cyber-attacks and unauthorized access when IoT devices gather and transfer sensitive data. IoT solutions must be both scalable and interoperable in order to handle an increasing number of devices and to enable smooth communication between various platforms and technologies.[15].

All things considered, the Internet of Things continues to transform a variety of sectors, including manufacturing, smart cities, healthcare, and agriculture. It is offering new tools that enable previously unheard-of degrees of operation control, optimization, and monitoring. Many IoT systems and information science addresses associated with IoT devices are online in order to provide regular services and tasks to end users and organizations.

Applications for the Internet of Things have multiplied dramatically in recent years, and predictions indicate that this trend will continue. In 2018, there were more than seven billion Internet of Things (IoT) devices online; by 2019, it's expected that fourteen billion will be linked.

In 2017, the residential sector had 664 million devices in use, making it one of the IoT applications with the largest number of deployed devices. Smart household appliances including lights, freezers, stoves, refrigerators, and locks are some of their uses. Proposals for several European countries have called for more IoT usage, or the "smart city." Currently, the most prominent factors driving the trend in industrial, agricultural, and health applications are automation, pricing power, and accuracy. Healthcare devices include, but are not limited to, energy-efficient, imaging-related, and patient monitoring technology (X-ray machines). [16].

B. CLOUD COMPUTING

Offering computer services via the internet and giving users access to a shared pool of programmable resources, including storage, software, and processing power, is known as cloud computing. This paradigm offers three key ideas and does away with the requirement for businesses to purchase and manage physical infrastructure: On-Demand Self-Service: Without assistance from the service provider, users are able to reserve and control computer resources, such as storage and server time, as needed.

The Internet of Things and cloud computing are the foundations of an expeditious and self-governing transition. At first, the nearly infinite cloud computing resources and capabilities may offset the limitations of IoT technology in terms of processing, storage, and communications. [17].

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The fundamental idea behind the Internet of Things is the connection of intelligent, self-aware nodes, or "things," in a dynamic, global network architecture. It is one of the most challenging technologies and promotes the possibility for ubiquitous and pervasive computing. Some refer to the Internet of Things (IoT) as a term for compact, adaptable, worldwide, and accountable objects with constrained processing and storage capacity as well

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as privacy, security, and accountability issues. However, cloud computing is a highly advanced technology that provides almost infinite processing and storage capacity, and it has addressed or significantly resolved the majority of Internet of Things problems. Therefore, it is expected that the present and future internet will be disrupted by a single IT paradigm, where Cloud and IoT are complementary technologies[18]. Since the inception of the fog idea, an interface between cloud and fog infrastructures has been required[19]. Additionally, [18] focuses on the communication between these entities; one of the major issues is the standardization of this interface [20].

C. EDGE COMPUTING

Edge computing is key components of distributed computing, aim to shift processing and data storage closer to the point of demand in order to speed up response times and conserve bandwidth. Their unique characteristics may either enhance or criticize one another, even when their objectives are the same. Edge computing handles data on devices at the network's edge, including smartphones and Internet of Things sensors, while fog computing uses an intermediary layer, like routers or gateway devices.

The combination of these two allows for a more efficient distribution of computational tasks through a layered computing architecture, increasing system reliability, performance, efficiency, and scalability. Building small, virtualized infrastructures in between base stations, radio network controllers, and other aggregation sites at the network edge is necessary for edge computing. While edge computing targets applications similar to fog computing, it has a different architecture and operating system (one operator maintains the infrastructure, excluding consumer devices). In order to deliver mobile edge services, such as location, information measure management, and radio network information, edge computing also makes use of telecom infrastructures[21].

Applications operating on edge devices may be effectively moved to fog devices, which have more resources and battery capacity, to extend the life of the edge devices' batteries. Research has shown that this strategy works.[22]. However, combining edge and fog computing presents further challenges. The growing amount of data processing at the edge creates security and privacy concerns, and it is challenging to manage resources between fog nodes and edge devices. Lack of standardization can lead to interoperability issues, and interdependency on a network connection might cause issues in some circumstances. Moreover, careful management is required for the energy consumption and sustainability of such a distributed network due to the complexity of the deployment and upkeep of these integrated systems. All things considered, edge computing increases processing power, which benefits fog computing, but it also presents additional difficulties that need to be overcome to fully utilize distributed computing systems.

D. FOG COMPUTING

With the introduction of real-time IoT devices, fog computing has become an attractive complement to cloud computing, providing cloud computing to the edge of the network to meet the strict latency limits and demanding processing requirements of these applications [22], [23]. A typical fog compute node is made up of many geographically dispersed fog nodes that are situated at the edge of the network and have varying resource provisioning, including bandwidth for communication, processing power, and storage bandwidth [24]. Fog computing provides low-latency communications and a dynamic environment by bridging edge devices and the cloud. With its ability to do calculations and store data fields, it is more akin to the edge in design and closer to the cloud in terms of resources.[25].

Online gaming, healthcare, and autonomous driving are just a few of the time-sensitive applications that fog computing, a cloud extension, occasionally lowers latency and helps. Local devices perform basic tasks, whereas huge cloud installations manage complex operations. Fog computing gathers virtual resources from several, dispersed, and independently controlled devices in the environment [21]. Networks and things may link thanks to the Internet of Things. An inventive idea called fog computing gives Internet of Things (IoT) devices (such sensors or embedded devices) networking, storage, and computing capabilities at the edge of the network. Benefiting mobile computing, fog computing quickly analyses vast volumes of data and controls billions of internet-connected devices.

Fog property, service quality, programming model and interface, machine offloading, scheduling, and accounting

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and tracking, information privacy, and resource and repair management are some of the core issues of fog computing. Fog computing uses a cost-effective cloud architecture to remove disruptions. Options for cloud assistance can be obtained by placing fog nodes close to the information source. Fog analyses and saves IoT data locally at IoT devices rather than sending it to the cloud, allowing fog to provide better services with faster reaction times. When it comes to enabling a variety of IoT applications in an IoT system with useful and trustworthy services, fog is the best option.[26].

Fog will be differentiated from cloud by the proximity of end users, allocation that is focused on a particular

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region, and provision for quality. Using fog significantly reduces the amount of network information measured in comparison to cloud computing. Every distinct piece of IoT data has to be transmitted to the cloud data center for cloud-based IoT data processing. The cost of data transfer increases as the volume of data to be examined climbs quickly, as it does with today's IoT. Information is stored and processed locally due to the fog, which lessens the usefulness of conveying such information. Processed data is uploaded to the cloud if it has to be kept for various analytical and historical purposes. As a result, consumers would profit from lower operational costs. Additionally, because the data is processed so near to the source, the end-user application runs extremely quickly, which is essential for preserving service quality over time and for mobile-to-mobile operations. As a consequence, IoT information services become more dependable, efficient, and consistent, and fog management services enhance customer satisfaction.[26].

While fog networking appears to be a workable way for working around some of the problems with cloud computing and current networks, there are still certain issues that need to be resolved. A distributed intelligent platform is the most important need for controlling networking, compute, and storage resources at the edge. It is difficult to choose an appropriate distribution strategy in fog networks due to the large range of processing power capacities of nodes, the unpredictability of job demands, resources available at fog nodes, and distribution options [27]. When choosing a distribution, node connection delays should also be taken into account since they may result in longer processing times [28]. As a result, there are many difficulties facing the fog computing paradigm [29].The response time of a fog node is used to determine whether it should execute a service request in its entirety, only some of it, or forward it to another fog. The waiting line volume and the service's demand travel time will be used to establish each fog's reaction time on a regular basis (lower latency is always preferred)[30]. Every fog in the network attempts to collect information about neighboring fogs and nodes. When a fog needs to transfer its responsibilities to another node, it maintains its connection to the nodes around it, modifies the computational function's value and the connection's state, and uses an algorithm to choose which fog to transfer that analyses the different fogs.

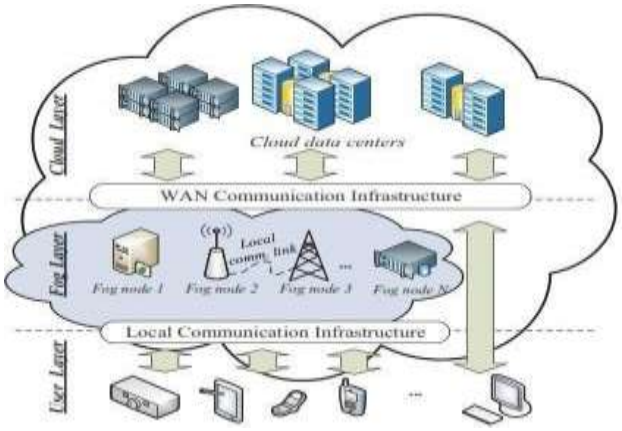

FIGURE 1. A three-layer fog computing architecture [22].

On the other hand, a server or other centralized device is used in the centralized architecture to map out connections between distributed fogs over a network, assess computational resources, and collect data from distributed fogs. Constantly send the nodes' communication connections to the central device. In order to submit its tasks, a node will interface with this centralized system. Based on nodes and network connections that follow the decision-making process, central devices choose the target fog for offloading and introduce the source node. Figure 1 depicts an architecture for three-layer fog computing.

E. *TASK OFFLOADING*

IoT and mobile devices have limited memory, processing power, battery life, and communication data measurement, making them unsuitable for executing large-scale programs. To address these issues and solve the current dilemma, mobile cloud computing was created. It is a paradigm for offloading computation from mobile devices to the cloud, freeing up resources that would otherwise be restricted to mobile devices. Offloading tasks

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to the cloud is limited to certain mobile apps due to a number of issues, such as a significant average access latency between users and remote clouds, excessive battery consumption, and a lack of local user data. In order to solve these issues, new strategies are being put forth, such as shifting mobile network traffic to complimentary networks like Wi-Fi, satellite-terrestrial networks, and edge cloud computing [31].

Evaluate the scenario in which a fog node requests and processes data processing from a thing before replying. A fog node may only process a portion of the payload before forwarding it to other fog nodes if it is overburdened with other requests. Fog node interactions can be simulated in two ways: A central node maintains the fog node

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offload contact in the first technique. Each fog node adheres to a protocol in order to give its neighbor's the most up-to-date information about its state. Second, each fog node keeps an up-to-date list of the top nodes for handling the offloaded workloads. [30].

FIGURE 2. Task offloading scopes related to fog computing.

Fog computing presents a number of issues for offloading, three of which are: how to split an application for offloading, how to create the optimal offloading scheme, and what kind of data is required for offloading choices. [13], [32], and [33].

Fog computing resource allocation and dynamic offloading provide several important issues, according to [23], Among these difficulties are the following:

1) The Trade-off between Power and Latency and System Dynamics: A fog system often has several layers, complex dynamic interactions between fog tiers and the cloud, and dynamic internal dynamics that are always shifting.

2) Efficient Decision-Making Process: Decision-making procedures have to be computationally efficient due to the presence of several heads. The inherent complexity of the problem, the continuing nature of job delivery, and the unpredictable nature of the traffic data frequently contribute to the difficulties.

3) Recognizing the Advantages of Predictive Offloading: Using predictive offloading to cut down on delays and enhance service quality is an essential component of online decision-making. Fog computing prediction offloading has several uses, but its fundamental limitations are still unknown.

The main area of study will be resource management, specifically as it relates to data storage and compute resources across various network fogs. Tasks may be offloaded by Internet of Things devices to fog nodes, which can then offload to clouds and other fog nodes. Due to the fog node's limited processing capacity, the jobs are re-offloaded to the nearby fog nodes or cloud [34], [35], [36], [37], which increased the costs associated with transmission and processing[37], [38]. The breadth of job offloading in fog computing is shown in Figure 2. We will outline and talk about relevant fog processing and offloading initiatives here. Several academics have taken an interest in the fog processing topic because of its importance and relevance.

F. SOFTWARE DEFINED NETWORK (SDN)

SDN is a cutting-edge networking idea that separates the data plane from the control plane [39]. This results in easier network design and maintenance as well as greater flexibility and agility. The fundamental idea behind SDN is to delegate network knowledge and decision-making to a logically centralized software-based controller, or control plane, while the data plane handles activities like traffic forwarding. Network programmability, which encourages network automation, and network management, which reduces operating costs by reducing administrative tasks, are only two of the numerous benefits of SDN. and network virtualization[40].

In Internet of Things applications, packet forwarding on the network side is when most device power consumption occurs[13], [41]. Poor-quality offloading caused by insufficient bandwidth and high latency will decrease device life and degrade offloading quality. To address this problem, SDN has developed as an auxiliary technology for network offloading[13], [42].

Traditional fog networks suffer from a number of issues, including centralized control, programmability, load balancing, and administration [43], [44], and [45], [45],. SDN allows fog nodes to collaborate effectively [2],

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[45], and [46],. SDN gives the network's programming more leeway by improving the rules in its centralized SDN controller from a network-wide viewpoint [47]. An SDN controller dynamically maintains the network by collecting data from each network device, modifying the load-balancing strategy, and monitoring network traffic [45], [48], [49]. To guarantee that the field of SDN-enabled networks offers a high level of service, the main responsibility of the SDN controller is to gather network data, such as resource consumption, device location and mobility, load metrics, and network information [13], [50], [51], [52], [53]. Because it employs the SDN controller in accordance with compute needs and network circumstances, the offloading service with SDN can detect

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changes in the network and choose the optimal offloading node for an overloaded fog node.

III RELATED SURVEYS

As was previously said, the ideal configuration won't be achieved by just moving and offloading data across Fogs; job management and resource availability must also be taken into account. However, most previous review studies have approached algorithms in a general way, not considering the nuances of the algorithm's ability to manage tasks and resources. By examining fog analysis algorithms based on the three main capabilities of task management, resource management, and task offloading, this study significantly advances the field of fog computing.

This research departs from previous studies that concentrated on machine learning applications, mobile Edge computing, or fog and Edge computing. Rather, it provides a thorough examination of SDN applications and network issues, particularly in relation to fog offloading. This in-depth analysis of network and SDN problems closes a gap in earlier studies that either overlooked or just touched on these topics in passing. It thereby contributes significantly to the topic. This method closes a critical gap in the literature, making our study an invaluable tool for understanding and advancing network management strategies in these emerging computing paradigms.

The major topics of surveys [39], [40], and [44] were mobile Edge computing, Edge computing, and SDN utilization in general. But they either ignored or just mentioned network issues in passing. However, this study fills a major gap in the current body of knowledge by providing a thorough examination of these subjects and by providing insightful viewpoints for improving network management strategies in fog computing. Task offloading in edge computing and the Internet of Things is examined in [54]. The study takes into account the advantages, variables that affect the choice to offload, and various offloading methods and algorithms. Machine learning and non-machine learning algorithms are used to categorize the algorithms, and offloading techniques including whole and partial offloading are covered.

The function and difficulties of machine learning and artificial intelligence algorithms for resource management in fog/edge computing environments were examined by the authors of [55]. References [56] looked at important journal articles about mobile computing and the changes in research directions, difficulties, and techniques. Additionally, the fundamental elements and architecture of edge computing are explained. A summary of offloading tactics in a fog environment may be found in reference [57]. The publications covered by them were published on or before November 2020, and so do not include the most recent findings in fog computing research.

The aims, optimization techniques, and appropriate fog computing algorithms for job offloading are outlined in general in [58], along with the formulation of the mathematical problem. In the cloud-IoT ecosystem, the article [35] discusses several offloading approaches and looks at aspects that affect whether to use the cloud or the fog for offloading. The writers also cover additional technologies that facilitate offloading, such virtualization, AI, and wireless connections. Reinforcement learning (RL) and deep reinforcement learning (DRL) algorithms' approaches to fog computing's offloading issues are reviewed in reference [59]. The fog computing offloading techniques are classified into three classes: value-based, policy-based, and hybrid-based algorithms.

Unlike previous review studies, we approached task management in this study from a variety of angles, including SDN technologies and network issues. This work is contrasted with other pertinent studies in this topic in Table 1.

The relevance of networks and SDN in fog computing is the main topic of this review paper, which also highlights how these technologies enhance resource management, scalability, and computation efficiency in both distributed and centralized computing systems. Fog computing is an extension of cloud computing that brings processing and storage closer to the edge of the network, necessitating the use of flexible and dependable network management tools. SDN plays a crucial role in this context since it provides dynamic network design and optimization that can adapt to the various and perhaps unpredictable requirements of fog computing environments.

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FIGURE 3. The system of interconnected components that constitutes fog offloading.

This study's thorough analysis of these concerns not only provides a comprehensive understanding of the difficulties and current solutions in this sector, but it also paves the way for future developments in SDN application and network management—two areas crucial to the evolution of fog computing technologies.

IV FOG OFFLOADING RELATED APPROACHES

Fog computing operates on a very low latency paradigm. Many scholars are interested in fog computing because of its increased responsiveness and scalability [14], [35], [46], [59], and [60]. In the literature review, several publications in the fog computing industry were examined. Based on the types of methods and system architectures used, these publications are categorized and contrasted into the following fog offloading topics.

a) Resource management
b) Task management
c) Task Offloading

[Figure 3]. Illustrates the interrelated elements that make up the fog computing architecture. The sections that follow will address pertinent topics. We assess the surveys that are currently accessible on this topic and highlight the importance of this study in this section before moving on to the articles related to the discussion.

a) RESOURCE MANAGEMENT

The fog network is made up of a variety of heterogeneous devices that may communicate, store data, and execute calculations. Resource management is a major difficulty because of the dynamic nature of fog networks and the mobility of the devices that make them up. This may improve fog network performance and maximize fog's processing capacity, given the significance of resource management. Since then, other initiatives to enhance the resource management of fog networks have been evaluated. Fog nodes are encouraged to unload incoming requests according to their load and processing capability using a fog-to-fog cooperation model described in [30]. The mathematical model of fog-to-fog cooperation that distributes tasks across cooperative fog nodes almost as efficiently as possible. Motive of the resource management is based on the Energy-Efficient Resource Management & Security and Privacy Considerations. To Ensuring secure communication, Quality of Service(QoS) Considerations and data protection in IoT-Fog-Cloud environments.

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TABLE 1. Related works.

Survey	Year	Area			Focus									Remarks
		Fog	Edge	Cloud	Task Offloading	Task management	Resource management	Metrics	Algorithm types	AI Discussion	Network Architecture	SDN Details	Simulation Tools	
[54]	2023		✓		✓		✓	✓	✓	✓			✓	Focus on Edge computing
[56]	2023		✓		✓			✓	✓	✓		✓	✓	Focus on mobile Edge computing
[55]	2023	✓	✓	✓	✓	✓	✓		✓	✓				Did not discuss network issues.
[57]	2022	✓			✓			✓	✓					Did not discuss network issues.
[58]	2022	✓			✓			✓	✓	✓	✓	✓		Just short refer to network architecture and SDN application in fog offloading
[59]	2023	✓			✓			✓		✓		✓	✓	Focus on Machine learning usage and just generally mentioned which articles use SDN or not.
[35]	2018	✓	✓	✓	✓			✓		✓				Didn't discuss network issues.
This survey	2023	✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	Network and SDN discussion.

In order to alleviate the latency issue with service requests received from objects, load balancing is encouraged by the Fog Resource Management Scheme (FRAMES). Its foundation is the idea of fog as a service, which suggests that every fog node is independent in terms of networking, processing, and storage. Fog node load balancing is supported by a rigorous mathematical model used in this work. The goal of the DLAEC technique, which was first presented in reference [61], is to improve service quality by leveraging deep learning while accounting for each edge system's network capacity and human resource availability. Through a three- step decision-making process, the DLAEC

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technique ensures the concurrent execution of the maximum number of deep learning tasks while optimizing resource use at the edge. Unlike earlier models that assign a fixed number of deep learning layers to edge systems, DLAEC independently assesses edge access and determines the appropriate number of deep learning layers for identification at both the edge and cloud nodes. This technique reduces the need for cloud transfers in an Edge Computing-based intelligent city environment, hence reducing network congestion and load on the cloud by allocating most deep

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learning activities to edge nodes. However, its main goal is to enhance the capacity of the edge nodes rather than minimize processing time delays. As a result, delays might arise from the maximizing, and more work could accrue on one node before being sent to other nodes.

Blaster is a federated structure designed to enhance the scalability and performance of data-intensive applications by routing packets throughout a distributed edge network [62]. It also has a cutting-edge path selection algorithm that uses Long Short-Term Memory (LSTM) to determine the optimal path. To maintain data flow capacity while enhancing communication across SDN controllers, this approach makes use of a Federated Learning (FL) paradigm. Regression is a customizable approach used by LSTM models, a type of recurrent neural network, to choose the best path during system creation.

The LSTM method is widely used to solve deep learning-based time series forecasting issues. It gives the LSTM a network graph and a traffic matrix. The output, which represents the anticipated future load on the input connections, may cause the SDN controller or application to change the route selection in a peak load forecast. Consequently, route selection is influenced by the network's condition.

The implementation of SDN to fog networks, namely P4/P4Runtime, is covered in Reference [63]. The industry claims that P4Runtime, a control-to-data plan interface, and P4, a data plan programming language, may help meet the needs of next-generation networks. It offers a special method for implementing SDN control plans that can manage SDN data plans integrated close to the main network as well as fog network SDN data plans. By handling cross-layer SDN data plans in this way, SDN controllers can allow some processes to be offloaded to the Edge. More specifically, Reference [64] develops Android apps that act as bridges between IoT devices and Edge/Fog/Cloud Computing ecosystems. These modules enable communication with several devices that serve as data sources. Additionally, they can be easily integrated with different Fog/Cloud frameworks, which allows them to be easily customized for a wide range of applications. This idea is limited to a single Android application and has limitations when it comes tosystem implementation.

In [65], a Blockchain-based data allocation method for Internet of Things systems was proposed. Decisions on data allocation are aided by a fuzzy logic-based data controller. Fog Bus enhanced a healthcare case study by utilizing blockchain-based cloud and fog technology. While network usage, energy consumption, blockchain storage, and data transmission latency have all decreased, security mechanism delays continue. A paradigm for managing applications in processioning fog infrastructures was created and suggested in [66], using Edge Proximity as its foundation. With its resource-aware approach, this paradigm arranges applications in a way that makes use of more fog processes to execute applications with high data loads and demanding resource requirements. It deals with bandwidth and application data flow issues. This strategy's disadvantage is that it can't function without a static fog management server. A fog computing model is used in [67], to provide mobility assistance recommendations for individuals with visual impairments. The acronym PEN (Phone + Embedded Board + Neural Computing Stick) was created by combining a mobile phone, a low- cost, low-power embedded board, and a neural computing stick in one device. Collectively, these three gadgets combine fog and cloud computing to offer a wide range of distributed capabilities. One limitation of the approach is that the system's mobility and flexibility are affected by a variety of hardware components.

Reference [68] uses the network traffic matrix to demonstrate SOSW in Mobile Edge Computing (MEC). The SOSW model uses a combination of singular-value decomposition (SVD), QR factoring, and column- pivoting linear algebra algorithms to determine the best locations for fog nodes. The constraint-based shortest path algorithm (CSPA), a heuristic-based traffic engineering solution that optimizes route calculation for job offload, is provided by SOSW with the help of ant colony optimization (ACO). The primary focus of this study is fog architecture, from its grid to its place in the network. Energy usage may also be reduced by placing fog to IoT devices so that data can be offloaded to fog faster. Energy usage may also be reduced by placing fog to IoT devices so that data can be offloaded to fog faster. Chooses the suggested transmission mode and the path via which the IoT device may upload its data to the fog by gathering data via SDN.

Research [69] uses average failure, downtime, or repair time to measure connection quality across time. Use a k-nearest (k-NNR) machine learning technique based on regression trained on a 5-month real-world network data set to assess the dependability of connections. Connection factors including link failure frequency, link usage rate, and data set failure frequency are taken into account by the k-NNR. Additionally, figure out the connection latency by adding the transfer delay package size. The internet of vehicles in MEC is covered in Research [70], which uses

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SDN to provide a method for resource allocation and communication improvement. By predicting offload decisions, it uses q-learning to improve the distribution of computing and communication resources. The shortcoming of this design is that it fails to take into account some environmental elements that are essential to the Internet of Vehicles system's correct operation. It employs a deep reinforcement learning approach in SDN networks with an emphasis on edge computing-based computing offloading and resource allocation to find the optimal offloading option, communication, computing resource allocation, and privacy protection design. Terminal vehicle real-time data is directly fed into MEC's processing apparatus, and flexible network control and centralized resource management are made possible by the integration of SDN and MEC.

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An adaptive computing optimization for edge computing and SDN has been presented in study [9] for industrial IoT resource management. Priority-based transmission with the least energy is a method for efficiently determining the resources of nearby devices and edge nodes for task processing. A user revocation system that preserves data integrity in a multi-user cloud environment with an effective architecture is given in [71]. Compared to existing models in the cloud environment, it is effective at lowering overhead; nevertheless, as the number of users rises, so does the time it takes to establish signatures and proofs. Applications that use on-board cameras for traffic monitoring or CCTV cameras for surveillance and event monitoring may generate massive amounts of data, particularly when video stream processing is used field [72]. The creation of dynamic workload placement techniques for latency-constrained stream processing requests is the main contribution of [20]. The organization of operators and resource allocation in a dispersed environment utilizing cloud and fog computing is the main focus. The fog-cloud interaction is used in computing infrastructure provisioning when there is a scarcity of network capacity. The objective of the algorithms is to maximize the percentage of requests that are successfully processed while meeting application delay constraints and making effective use of the available resources. [73] presents the Set-based Differential Evolutionary algorithm for energy-efficient resource management in IoT-based smart cities, utilizing SDN and fog computing, in order to balance quality of service and energy usage in large-scale networks. According to [74], each fog node's processing power and bandwidth (communication capacity) characterize it in the resource allocation strategy for fog computing based on SDN. The process starts with selecting an admin node, after which, based on the fog node's capacity, the nodes in the task graph are mapped to it. After this mapping, a management module maintains the fog node's capacity, from which the allocated resources are deducted.

b. TASK MANAGEMENT

Various tasks are allocated to the fog for analysis and processing by utilizing fog in the processing of information connected to IoT devices in many industries, including media, transportation, health, and other areas of intelligent cities. There are several jobs accessible, from location and navigation for analysis and processing to large amounts of data and films captured by cameras and sensors. The majority of these devices and data require instantaneous responses and real-time fog data processing. Consequently, for jobs that have a deadline, efficient management of a variety of tasks is essential. A number of studies—some of which are included here—have created methods and algorithms for task management in fog networks.

Task scheduling and dynamic resource utilization were recommended by Tuli et al. to enhance quality in hybrid cloud settings for data-intensive applications operating in a centralized shared file system. It evaluates data file type, data transfer time, network speed, and data proximity for applications that require large amounts of data while planning and dynamically provisioning public cloud resources. This method may enhance service quality and reduce job placement in a hybrid cloud environment by reducing the amount of shared file transfers between nodes. As of right now, this method is incompatible with edge and fog computing because it was created for cloud computing [75].

[76] provides a low-weight distributed method for monitoring and self-organizing surveillance in fog networks. In addition to detecting accessible IoT devices, Fog Mon analyses the hardware resources (CPU, RAM, HDD) of the access nodes and the end-to-end network quality of service (latency and bandwidth). When choosing leader nodes, it doesn't consider bandwidth or storage.

A Blockchain-based fog network's privacy-aware job scheduling strategy is presented in reference [77]. An efficient method for job scheduling is the Ant Colony Optimization (ACO) methodology. It decreases work processing time on cloud VRs and maintains end-user device anonymity. Take into account the needs from edge scheduling to minimize application delay [78]. By analyzing the network, computation, and reliability elements of edge nodes, a score-based edge service scheduling strategy drastically reduces network latency and service time for live video streaming services. It limits the centralized optimization process, which is the drawback. In a fog setting, a fuzzy evolutionary organizer is offered for the multi-target resource allocation [79].

The approach meets the objectives of robustness and agreement index as task processing time, inter-task communication information, and task In the context of mobile edge processing, machine learning is used by

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distributed job scheduling algorithms, distributed device coordination, and [80]. This model makes use of the distributed ADMM optimization technique and Stackelberg game theory. This method can quickly and perfectly converge to intelligent edge processing. One big problem with the design is that it only affects one MEC server; if there are other MEC servers, the scheduling method can be different. A paradigm for secure data processing and storing at the edge and in the cloud was introduced in [81]. This method encrypts data in transit for privacy. Tasks in this design are divided into little pieces and sent between edge nodes for execution; in the event that a node cannot handle the job, it moves to its side nodes and so on, until a suitable node is found and completed.

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A paradigm for secure data processing and storing at the edge and in the cloud was introduced in [81]. This method encrypts data in transit for privacy. Tasks in this design are divided into smaller units and sent between edge nodes for execution; in the event that a node cannot handle the job, it moves to one of its side nodes, and so on, until a suitable node is found and completed. This system does not take into account the node's resources or capacity; rather, it just sends it for processing based on its proximity. It can stop working and have to switch to another node. When system needs rise, this data transfer and undefinable mobility might result in significant data delay problems.

Effective Task Scheduling Algorithms for IoT-Fog-Cloud Systems

Some of the factors influencing the choice of a scheduling algorithm for an IoT-Fog-Cloud system are scale, real time constraint efficiency, and heterogeneity of resources in an IoT- Fog-Cloud system. A Few of Algorithms are below.

1. Heuristic Algorithms:

I. First-Come-First-Serve (FCFS)
II. Shortest Job First (SJF)
III. Longest Remaining Time First (LRFT)

2. Metaheuristic Algorithms:

I. *Genetic Algorithm (GA):* Inspired by natural selection, GA can find optimal solutions by iteratively improving a population of candidate solutions.
II. *Particle Swarm Optimization (PSO):* Inspired by the social behavior of birds flocking, PSO can explore the solution space efficiently.
III. *Ant Colony Optimization (ACO):* Inspired by the foraging behavior of ants, ACO can find optimal paths in complex environments.

3. Reinforcement Learning-Based Algorithms:

I. *Q-learning:* An algorithm that learns the optimal action to take in a given state by maximizing future rewards.
II. *Deep Q-Networks (DQN):* A deep learning approach that combines Q-learning with neural networks to handle complex environments.

4. Hybrid Approaches:

II. *Combining Heuristic and Metaheuristic Algorithms:* Combining the simplicity of heuristic algorithms with the optimization power of metaheuristic algorithms.
III. *Combining Reinforcement Learning with Other Techniques:* Integrating reinforcement learning with other techniques like model-based optimization or supervised learning.

Key Considerations for Algorithm Selection:

- *Latency:* Algorithms that prioritize tasks with shorter deadlines or higher importance can help minimize latency.
- *Energy Efficiency:* Algorithms that consider the energy consumption of devices and network links can help optimize energy usage.
- *Resource Utilization:* Algorithms that effectively allocate resources to tasks can improve overall system performance.
- *Scalability:* Algorithms that can handle large-scale systems with many devices and tasks are essential.
- *Fault Tolerance:* Algorithms that can recover from failures and re-schedule the tasks can improve system reliability & performance.

By carefully considering these factors and selecting the appropriate algorithm, it is possible to achieve efficient

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and reliable task scheduling in IoT-Fog-Cloud systems.

c. *TASK OFFLOADING*

Because there are many different types of fog in the network, it's critical to identify which fog should be offloaded when it becomes too much or when IoT devices require an external processor. Thus, the decision of the destination country selection affects the processing time, data delay, and deadline. On the other hand, another crucial offloading activity is the destination fog's dependability in doing the necessary processing without any mistakes. To establish the fog's credibility, variables like the destination fog's processing power, memory, and acceptable and

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sufficient connection for data transmission can all be used. At this point, choosing the best fog and obtaining fog information from the network require careful planning and execution. Numerous methods have been devised to get fog information inside the network, which may be classified into two types: distributed and centralized. SDN technology is integrated into most centralized algorithms due to its benefits over SDN networks.

i. DECENTRALIZE TASK OFFLOADING

Certain fog offloading techniques are built without SDN technologies or on distributed architecture, as was covered in the previous section. In order to reduce task execution latency for heterogeneous fog nodes, the matching theory-based distributed computing offloading framework MATO was presented in the research field[82]. The framework for distributed processing on an Akka-based fog network was improved by reference [14]. One of the most complete and popular actor modelling packages for the Java Virtual Machine (JVM) is the Akka toolkit. A task offloading technique using parked automobiles on the side of the road as a compute offload point is presented by [37] in the context of the Fog IoV network. Field [83] treats the competition between IoT devices for fog nodes as a game. The finite improvement feature of the Weighted Potential Game illustrates the Nash equilibrium. Dec Chain is a blockchain-based secure edge computing system that does away with the need for a reliable third party on the network [84]. Issues like removing a reliable network failure point when a third party is no longer accessible have been dealt with. Among the difficulties it encounters is incorporating blockchain into the edge processing environment. Services providers or edge servers oversee network tasks. As can be observed, service providers manage jobs according to a predetermined structure without taking edge nodes' ability to divide tasks into smaller ones into account. Multi-hop vehicular systems are the focus of reference [85] in order to provide the best possibilities for carrying out tasks locally or remotely. Effective route selection and fog node assignment can lower average service latency and energy usage for compute offloading. When there are fewer than 50 cars, it performs worse than other frameworks; however, when the number of cars increases, it performs better than other frameworks. Research [86] integrates optimization strategies for resource allocation and job offloading in 5G network technologies based on edge computing by managing the system's energy usage. Even greater latency performance is possible with this approach, which would improve end-user happiness and the operation of 5G mobile communication networks. The proposed method solely takes into account the total processing time delay of all tasks inside the system, without taking into account the order in which computing jobs are accomplished.

In [87], a paradigm for data quality-aware task offloading in mobile edge computing is described, inspired by the Optimal Stopping Theory (OST). When mobile nodes require few resources and make tiny, autonomous decisions inside the MEC environment, the OST-based approach is appropriate. Without taking into account the context of other mobile nodes, each offloading mobile node will execute the models in a single configuration.

A GPU-based embedded edge server is used to demonstrate the architecture and workload offloading based on neural network server [88]. The compute difference between the edge and the central cloud is the basis for the offloading technique that is given. Building it is very expensive since it relies on GPU resources at the periphery of the network. According to [22], if a network node is too busy, it can use a cooperative method to upload all or some of its duties to other fogs. This method, which uses a multiplier distributed alternating direction strategy, enhances fog dispersal. The Quality of Experience (QoE) and power efficiency of fog nodes are the main objectives of this model, along with the trade-off between the two. A multi-layered fog computing system for traffic prediction-based dynamic resource allocation and offloading is provided by Reference [23].

It characterized the issue as one of stochastic network optimization and provided a workaround that keeps queue stability while consuming less electricity. The idea's drawback is that if prediction errors occurred, power consumption and total queue backlog sizes would become inefficient. In order to offload task data concurrently [36], recommends working vertically with a distant cloud and horizontally with several fog nodes. It considered the time it took to transport data and deliver services, as well as the latency in communication between end users and the fog. It also took into account waiting periods at the fog node connected to queuing, as well as local computation times.

The focus of the writers [89] was on offloading responsibilities from a device to fog and fog-to-fog. This model looks at the best scheduling strategy for the two queues, such as low and high priority in a fog node. While further offloaded jobs that are aware of deadlines may be completed, the stability of both queues is simultaneously

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maintained. Two methods make up the suggested strategy: A priority-aware scheduling technique based on the queue state of the fog nodes is constructed through the collaboration of fog computing and the Lyapunov drift plus-penalty method. The results of the simulation indicate that, with the same resource configuration, this technique might guarantee that more jobs are completed within the given time frame.

ii. CENTRALIZED TASK OFFLOADING

Centralized task offloading is divided into two categories: those that use a SDN based Task Offloading and those that don't use.

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a: SDN-BASED TASK OFFLOADING

In this section, we look at and assess SDN-based fog offloading techniques. The majority of these methods are designed to dump data to fog from mobile devices or the Internet of Things. According to by [2], a dynamic offloading service between fog nodes is recommended for an SDN-based fog computing system. The objectives of utilizing SDN technology are to identify the appropriate offloading node and enable the offloading path by offering an end-to-end bandwidth guarantee. The suggested approach uses real- time network status and fog node information to select an optimal offloading node. Additionally, a few network settings may be employed to optimize the end-to-end path selection route and guarantee bandwidth across the offloading path.

The controller may select the best course of action according on the reward function thanks to Q-learning offloading choice [29]. The controller may tailor their incentive function in this method in accordance with the intended performance. As a consequence, the proposed Q-learning-based offloading option may predict the future offloading performance, leading to exceptional system performance. Additionally, because the reinforcement learning-based decision-making process is on-demand and enables the controller to create a reward function based on the needed performance, the proposed method is compatible with SDN architecture. The problem of multi-hop task offloading is solved by [90] through the use of integer linear programming (ILP). Since the viable set is non-convex, a greedy heuristic-based method is the most effective way to tackle the problem. The greedy method includes latency, energy costs, multi-hop routes, and dynamic network parameters like SDN rule capacity and link usage. The input 1) The best choice is to compute a task locally or remotely; 2) The best choice is to pick a fog node; and 3) The best choice is to select an offloading route. The suggested method achieves good performance with fewer fog devices. SDN techniques are applied in [31] to improve load balancing and resource management over a network of Cloudlets. The job distribution that is offloaded from mobile devices is balanced while maximizing resource use, and this is achieved through the use of an optimization model known as mixed-integer linear programming (MILP). The problem of a balanced distribution of incoming requests throughout the Cloudlet network is described, considering available resources, a job with a limited deadline, and delays in both computation and communications. Try using Floodlight as the SDN controller and Mininet-WIFI as the Mininet WIFI to mimic the suggested design. The resources that are accessible and the customer wants are the method's constraints. In order to enhance the offloading option selection process, Reference [11] developed a mix offloading structure that ensures the required deadline for the infeasible request, as well as the average processing cost of the fog node, while also choosing the best offload node. It uses the Binary Linear Programming (BLP) mathematical technique to decide which data should be offloaded. The SDN controller of the fog node will determine the best offload target fog by comparing the response times of all available parked and moving cars with the necessary deadline of the given request. A four-layer SDN-based network paradigm for the Industrial Internet of Things (IoT) is proposed in Reference [91]. All potential compute offloading sites indirectly manage the production equipment to tasks at the nearest power steering device based on the requirements of various controllers. SDN controllers manage the routing of data transmission, and processing is provided by lightweight containers like Docker and Micro. By working together amongst fog computing nodes, reference [92] provides an algorithm in fog system resource management to guarantee each task's unique service quality and optimize resource usage. To optimize tasks within time constraints, develop a common heterogeneous task download and resource technique using Deep Recurrent Reinforcement Learning. As per the buffer and resource status, the suggested approach could transfer their responsibilities to neighboring nodes. Ensure that resource allocation slices with different priorities are not unfairly distributed, leading to an increased average success rate. Based on the load assessment of cloudlets in multi-cloudlet networks, [93] determines which cloudlet should be used to offload a certain job from mobile devices. Additionally, in wireless mobile networks, this system uses SDN to load balance across cloudlets. It also recommends an admission control system to restrict the number of task assignment requests that are accepted in the event that the network is overloaded. The authors of [94] suggested a software-defined vehicle network offloading method while taking into account the multi-hop route and the effect of vehicle mobility. In order to minimize operational and capital expenses, it offered an ILP-based optimization problem for figuring out how many fog nodes are best for a particular network. Research looks on the dynamic offloading for soft real-time workloads in an SDN-based fog architecture [95]. The influence of intra- and inter-fog cluster analysis on the cost of missing the deadline has been highlighted by preliminary results.

The offloading issue in IoV systems based on SDN and fog computing is examined in study [50]. It suggests an energy-aware dynamic offloading method to increase the IoV system's battery life and allow it to run additional

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programs. Costs are associated with calculating the system model and transferring applications, and the optimization method known as heuristic searching occasionally need help to arrive at the optimal solution. However, the experiment scenario did not include testing on a real-world IoV edge network.

Direct execution of resource and system functions on the programmable data plane is made possible by the Programming Protocol-independent Packet Processors (P4) framework, which includes the recently announced P4. This feature allows some functions to be transferred from the controller to dedicated hardware, such P4 switches. Programmers are able to define packet behavior on the data plane with P4 [13]. A P4-assisted task- offloading

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technique for fog-based IoT networks is proposed by [13]in order to maximize the effectiveness of task-offloading solutions and lower computational overhead and latency in IoT networks. The offloaded job from the IoT device will, however, be done instantly if the fog server has the required resources; if not, it will be placed in a queue and wait until those resources are available in the fog. This leads to a decrease in system efficiency and asmall lag in task processing.

b: OFFLOADING TASK CENTRALIZED (WITHOUT SDN)

Centralized architectures are provided by certain study without the use of SDN. Fog bus is a blockchain-based platform that combines cloud, edge, fog, and Internet of Things communications through data encryption and user identity, according to the authors of in [96]. Tasks are assigned to other fog nodes known as workers by a fog known as the master fog. The master fog and worker fog nodes are selected at random by the algorithm. System flexibility is decreased when massive amounts of data and activities are involved. To determine the final processing node delays and system performance, data must be modified and transmitted from numerous nodes. This intricate method makes use of a basic blockchain process that the system administrator may start or stop. When compared to the inactive state, the system latency increases as soon as the blockchain is started. An affordable compute offloading design for usage in industrial networks may be found in Reference in [97]. It operates on the basis of a fog federation, where a master fog controller controls the data flow and traffic from IIoT sensors to the different fog nodes. In addition, it employed a controller-based device adaptation method and a policy-based reinforcement learning technique with a Q-learning algorithm to efficiently manage emergency- based service requests and direct them in the direction of the fog devices via the shortest path. A hierarchical model for the cloud, fog, and Internet of Things is introduced in reference [98]. Distributed task execution might allow for extremely scalable Internet of Things applications and regulate the world's energy use. Examine the effectiveness of a three-tier architecture as a proof of concept by taking into account the processing requirements of several Internet of Things applications, including text, multimedia, geolocation, and medical. Using real-world datasets, it assesses three use cases: fog-only, cloud-only, and fog-cloud cooperative.

V. EVALUATION AND COMPARATIVE ANALYSIS

This section investigated at and contrasted the offloading fog research that was presented in the offloading section. An overview of the algorithms under study is given in Table 2, which also includes information on the methodology, the sort of protocol and SDN used, the metrics used, the kind of simulator used, the area of the algorithms, and their salient characteristics and limitations. The bulk of the studies and algorithms deal with IoT to fog offloading, as the analysis of the articles demonstrates, while a few techniques for offloading from one fog to another fog have been covered. Examining fog-to-fog techniques with SDN might enhance a number of things. One of the primary roadblocks in the decision-making process and destination is fog selection, a decision-making technique that requires a high real-time processing need. However, a major challenge faced by most of these algorithms is lowering system efficiency through an increase in fog and network traffic. Additionally, while selecting the connection for offloading to the destination, the network data plan could be more useful. When selecting a fog, connection traffic and data plans also need to be properly taken into account. Failure to do so might result in increased delay in data transfer to the target fog and the failure of the offloading process in the event of a link failure. A synopsis of the comparative analysis and significant flaws in the present methods is provided below:

A. FOG OFFLOADING ALGORITHM TYPES

We can classify job offloading options as shown in Figure 4 fog computing uses a range of task-offloading mechanisms based on the evaluated literature. Machine learning adapts to changing circumstances by making decisions based on data, but it may require a large amount of training data and processing power. Although mathematical optimization is difficult and less flexible, it can produce nearly optimum solutions when properly represented. Although heuristic methods are straightforward and effective, they are not perfect. Offloading using blockchain technology increases decentralization and trust, but it may also come with additional costs. Application needs and resource availability dictate the selection process, which strikes a balance between trust, complexity, and flexibility. In summary, different fog computing job offloading mechanisms offer advantages and disadvantages depending on the objectives of the application. The choice should consider the particular requirements of the application in addition to the available resources in order to strike a balance between flexibility and

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affordability. Fog computing, in conjunction with blockchain technology, AI algorithms, and mathematical algorithms, has produced a variety of intricate solutions for resource and data management. These methods are important in the field of fog computing because they each have unique benefits and limitations. Fog offloading frequently makes use of mathematical approaches for a variety of goals, including task offloading, load balancing, energy management, and delay minimization. Their methodical and targeted approach provides useful solutions for optimization challenges including balancing computing workloads and reducing energy usage. However, because their efficacy is largely dependent on the specifics of each case and the algorithm's design, they seldom yield the best results and may need a large amount of processing power. Due to its distributed nature and cheap

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processing cost methodology, matching-based strategies provide potential benefits, even though optimization and game theory based solutions are viable in some limited scenarios [82]. Nonetheless, fog computing employs AI algorithms for adaptive resource management [92], dynamic decision-making [29], and predictive analytics [97]. They are very adaptable and have a strong propensity for learning, which makes them ideal for changing and unforeseen situations. AI's ability to automate complex decision-making processes and anticipate future requirements is its main benefit. But it also comes with drawbacks, such the requirement for large datasets, the complexity of training and using models, and even the opaqueness of decision-making procedures. A more recent development in the realm of fog computing, blockchain technology [30], offers a definite advantage in terms of security and data integrity maintenance. Because of its distributed and tamper-evident qualities, blockchain ensures a high level of security by permitting transparent and distributed transactions and communications inside fog networks. In distant computer systems, where data quality and dependability are critical, this capacity is indispensable.

TABLE 2. Summary of task offloading algorithms.

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Article	Method	SDN	Metrics						Remarks	Simulation Environments	Area	Cloud offload
			Delay	Energy	Deadline	Transmission	Processing cost	Number of Tasks				
2]	SDN based Algorithm for ranking hops	Yes	✓			✓			Linear algorithm for decision making and search hops in controller. The number of recourses of nearest selected nodes are lower than previous framework. By increasing the number of Fogs, the delay time increased and throughput came down	Mininet	Fog to fog	No
[14]	Establishing a neural network training environment for a fog network on top of an Akka-based distributed processing platform.	No	✓						Has relatively lower accuracy and lower latency compared to centralized model.	Real Testbed	Fog to fog	No
[37]	The offloaded tasks can be assigned to the best RSPVs using a mathematical model created by the CPLEX optimization software.	No			✓	✓	✓	✓	Fog act as offload coordinator among IoV and Roadside Parked Vehicles (RSPV)	-	Fog to fog	Yes
[82]	Distributed task offloading architecture based on matching theory	No	✓						To keep track of other neighbor fogs available resources, fog must regularly gather data from its surrounding neighborhoods. The expense of communicating through each fog is quite significant.	Simulator not mentioned	IoT to fog and Fog to Fog	Yes
[83]	Distributed task offloading algorithm from IoT devices to fogs using Game theory.	No	✓	✓		✓	✓		Have relatively higher average processing time compared to other baselines.	Python	IoT to fog	Yes
[95]	Real-time tasks being dynamically offloaded in an SDN-based fog architecture	Yes			✓			✓	focus on the consequences of the proposed intra- and inter-fog cluster offloading	-	Fog to fog	Yes
[18]	Employing P4 to assist with task offloading in Fog-based IoT networks.	Yes	✓					✓	When measured in accordance with various server load status control intervals, TOS-SDN performs better initially, but as the load status control interval grows, it eventually equals TOS-P4 in terms of average waiting times.	Real Testbed	IoT to fog	No
[20]	Placing operators and allocating resources in a distributed system employing cloud and fog computing	Yes				✓		✓	Resource needs and latency thresholds, which are aspects of stream processing requests, are not taken into consideration.	Mininet	Fog to cloud	Yes
[50]	Fog offloading using SDN in IoV systems.	Yes	✓	✓		✓	✓	✓	The ideal answer cannot always be found using a heuristic search optimization technique.	Real Testbed	IoV to fog	Yes
[29]	Use reinforcement learning technique that Q-learning was combined with the greedy algorithm	Yes	✓						Limit number of fog node	Realistic network	Fog to fog	No
[90]	Greedy-heuristic-based strategy	Yes	✓	✓					Limit to IoT to fog and with fewer fog devices, the suggested strategy can obtain good performance	Mininet	IoT to fog	No
[68]	Employ column pivoting linear algebra with singular-value decomposition and QR factorization.	Yes	✓	✓				✓	Limit to static IoT device.	Mininet	IoT to fog (MEC)	No
[11]	Use mathematical model named Binary Linear Programming (BLP)	Yes	✓		✓		✓		Don't consider energy consumption	MATLAB	IoT to fog	Yes
[70]	Use Q-learning for allocation of communication and computing resources	Yes	✓			✓	✓		The various environmental variables of the Internet of Vehicles system's real functioning are not considered.	MATLAB	IoT to Edge (MEC)	No
[91]	For IoT scenario suggest a four-layer SDN-based network design.	Yes	✓	✓					Limit devices are including in the real network.	Real Network	IoT to fog	Yes
[69]	Link reliability may be improved by using machine learning (ML) and multi objective optimization (MOO) methods.	Yes				✓			Limit to link optimization	Mininet	IoT to fog	No
[92]	Combines recurrent layer with deep Q-learning.	Yes	✓						Simulated Network consist of 5 fog nodes only	Python TensorFlow	IoT to fog	Yes
[93]	Load balancing between cloudlet by using SDN, also task assignment in wireless cloudlets network	Yes	✓						Limited to mobile devices task offloading in multi-cloudlet networks	Numerical simulations	IoT to fog	No
[94]	Use a greedy-heuristic approach	Yes	✓	✓			✓	✓	vehicle-to-vehicle communication are not considered, and fog determines statically.	python-based simulations	IoT to fog	No
[30]	Formulated a formal mathematical model of fog-to-fog.	No	✓					✓	depend on Web based app, consider nodes are reachable always.	MATLAB	Fog-to-fog	No
[96]	Blockchain base algorithm	No	✓	✓		✓		✓	Depend on special architecture for implementation. Algorithm design is limited to cloud PaaS Infrastructure	Aneka	IoT to fog	Yes
[22]	A distributed optimization algorithm to share workload among fog nodes based on distributed alternating direction method of multipliers	No		✓					Complexity of algorithm for each fog	Not specified	Fog to fog	No
[23]	dynamic offloading and resource allocation in fog network with traffic prediction (stochastic network optimization)	No		✓					In the event of forecast mistakes there will be a decline in efficiency in both overall queue backlog sizes and power usage.	Simulation by Python	Fog-to-fog	Yes
[36]	The optimization issue expressed as a quadratically constrained quadratic programming problem that is solved via semidefinite relaxation.	No	✓			✓	✓		Low efficiency and high complexity.	Monte Carlo and MATLAB	Fog-to-fog	Yes
[88]	Priority-aware scheduling strategy is defined using the Lyapunov drift-plus-penalty algorithm, and fog cooperation under the task scheduling strategy.	No				✓	✓		Limited to task scheduling in fog	Monte Carlo simulations	Fog to fog	No
[97]	Master fog controller uses a probabilistic method to select the most capable fog device for offloading and uses the Q learning to determine the effective route to reach the target fog.	No	✓	✓			✓	✓	Depend on central fog controller that in terms of unavailability of that the system can't operate.	Python	IoT to fog	No
[98]	Mathematical model with three layers to process tasks arriving from IoT and determine to offload them to a fog or cloud.	No	✓	✓					Global Gateway act as broker to distribute task among IoT, fog and clouds. Failure in Global Gateway may make all the system process down.	SFogSim	IoT to fog and IoT to cloud	Yes
[73]	Addresses leveraging SDN for energy-efficient fog resource management.	Yes		✓					Neglect to handle security issues related to virtual machine migrations, particularly when sensitive data and vital services are at stake in smart city environments.	CloudSim and iFogSim	Fog to fog	Yes
[74]	Task offloading in SDN fog environment	Yes	✓				✓		The depth first search (DFS) technique for pathfinding among fog nodes lacks optimality in finding the shortest route and may require excessive memory in deep or infinite branches.	Python	Fog to fog	Yes

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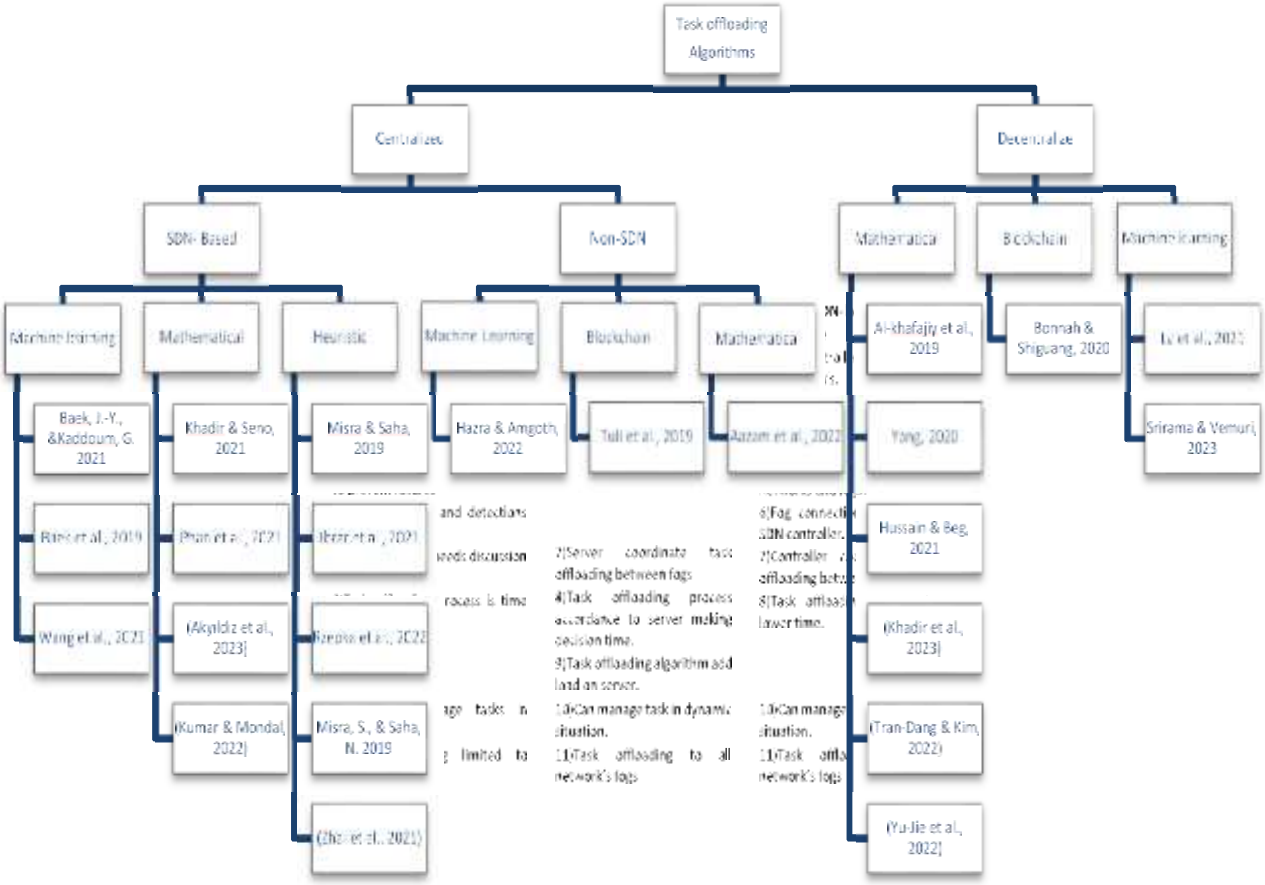

FIGURE 4. Task offloading algorithm types.

However, blockchain is subject to several limitations. Issues with scalability emerge, particularly in the face of increased transaction volumes, and maintaining a blockchain network requires substantial resources. Moreover, the integration of blockchain technology into existing systems presents a unique set of challenges. Ultimately, certain factors like security, adaptability, resource efficiency, and environment complexity dictate which mathematical algorithms, artificial intelligence, and blockchain are used in fog computing. In certain cases, mathematical algorithms provide exact and effective answers, but artificial intelligence (AI) offers flexibility and predictive power. Furthermore, blockchain guarantees unparalleled integrity and security for distributed systems. In the complicated world of fog computing, a hybrid approach that leverages the benefits of these disparate technologies is often the most effective solution. In order to successfully deploy SDN-based network characteristics, route selection and decision selection algorithms must be studied and improved in order to address the obstacles and challenges associated with fog-to- fog offloading. Many of the methods that have been seen now use machine learning, deep learning, or artificial intelligence (AI) algorithms in fog computing techniques as a result of the advancements in these fields. These techniques have greatly improved fog offloading forecasts, but because of their high computational cost and the network's restricted processing capacity in the fog, using them comes with a number of challenges. Problems include: While computational resources are abundant for deep learning, machine learning, and artificial intelligence algorithms, they are scarce in fog. However, the complexity of fog computations is increased by the amount of bandwidth and communication load needed to transfer data between devices.

B. FOG COMPUTING ARCHITECTURE

Different strategies have different implications on fog computing algorithms: centralized, distributed, and SDN. While algorithms provide theoretically optimal solutions for task offloading in centralized systems with a global view of resources and network conditions, they also provide a single point of failure and scalability issues. Although

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distributed approaches are less globally optimal, they are more resilient since they distribute decision-making across fog nodes based on local data and collaboration. The effectiveness of SDN-based designs relies on network programming and algorithm integration, but they offer dynamic network management, enhancing algorithm flexibility and responsiveness to real-time network changes. Table 3 displays the properties of both centralized and distributed task offloading strategies in comparison. As can be observed, centralized networks—especially SDN—offer a number of advantages in spite of their drawbacks and lack of flexibility. With high capabilities and minimal burdens on fog nodes, centralized management utilizing controllers in the SDN network makes it easier to regulate and manage the computing resources of various fogs in the network. Many of the current solutions would

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not be able to adjust appropriately in dynamic fog situations when node availability, network conditions, and workload fluctuate. Strategies that lack flexibility may not perform as effectively as they might in evolving situations, which could have an impact on the system's overall effectiveness.

TABLE 3. Task offloading networking schemes comparison.

Distributed Task offloading	Centralized Task offloading	
	Tags	
2)Hard to manage		
	6)Fog connection monitor by server	
9)Task offloading algorithm add load on nodes		9)Task offloading algorithm add load on controller

It is crucial to mitigate the negative effects of a highly dynamic nature, such as the unavailability of fog devices and the inability to enable end device mobility. SDN's centralized, intelligent network view and control makes it a viable network architecture for fog computing [95]. An outline of the advantages and restrictions of SDN architecture for fog offloading is provided below, along with a discussion of SDN protocols:

1) SDN ADVANTAGES

Real-time centralized network management based on the state of the network at that moment and user-defined rules is made possible by SDN, which makes it possible to acquire network state instantly. Moreover, fog networks need this to offload jobs instantly and without delay. This leads to benefits in network performance enhancement and optimization [99].

When adding new hardware to an already-existing network, configuration is very important for network administrators. The complexity arises from the variety of network device manufacturers and their configuration interfaces, which lead to time-consuming and prone to error manual setup processes. It will take a lot of troubleshooting to fix these issues. SDN provides centralized and automated configuration via software control by combining the control plane across many network devices. This facilitates network management and enables adaptive optimization based on the specific conditions of the network at that moment. By enhancing their ability to swiftly and simply manage and modify fog devices to allow new applications and services, fog computing networks are significantly impacted by the usage of SDN in network setup. Flow rules may be easily and optimally changed using SDN, taking into account the current state of the network and demand [100]. SDN simplifies network resource management in fog computing, improves load balancing, job offloading, latency optimization, and has a substantial influence on mathematical algorithms [2], [11], [13], [20], [31], [50], [68], [90], [91], [95]. These algorithms are enabled by the dynamic resource allocation to optimize distribution according to predefined criteria and real-time data. Furthermore, SDN's flexibility promotes algorithmic agility in the face of network changes by enabling quick modifications. SDN enables effective data management and routing, which is essential for decision-making processes in the field of AI algorithms [20], [29], [69], [70], and [92]. Its flexibility allows for real-time resource optimization, enhancing the scalability and performance of AI systems on a variety of networks. Moreover, SDN supports the dispersed, secure character of transactions in blockchain integration [30], improving network

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security and maximizing resource utilization. Additionally, it promotes enhanced interoperability across fog computing nodes, which makes the integration of blockchain systems easier.

2) SDN CHALLENGES

Traditional SDN-based handover has been hampered by previous research in several ways, resulting in higher latency and packet losses due to centralized control. The primary SDN controller is overworked due to the rise of mobility vehicles, which makes it challenging to meet quality of service standards and results in frequent handover issues. Comprehensive simulation studies are necessary to further restrict their effectiveness in high-mobility situations.

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Article [10] provides a Vehicular ad-hoc networks (VANET) architecture that closes this gap by using fog computing and SDN technologies to move changeover operations to the network's edge. By decentralizing changeover management via the use of zone SDN controllers and fog computing vehicles, this flexible solution meets the demands of highly mobile and data-intensive services while reducing the strain on core networks. There are major challenges in integrating SDN with fog computing that need to be overcome. Among them include ensuring that a centralized control system can manage a large number of fog devices and remain flexible; enhancing security and privacy inside the system; ensuring that various technologies operate together; and optimizing resource allocation to divide the burden effectively. The crucial issues of minimizing latency while optimizing traffic flows, improving the energy efficiency of SDN controllers and fog nodes, guaranteeing dependability and consistent Quality of Service in dynamic fog environments, and smoothly incorporating SDN into existing network infrastructures must also be addressed. Due to the low number of SDN controllers employed in the evaluated study, multi-controller scenario difficulties in huge networks must be taken into account and explained. In order to guarantee that SDN is effectively used in fog computing, it is imperative that these deficiencies be addressed. Industry collaboration and continuous innovation will be required for this.

3) SDN PROTOCOLS

Wide-ranging compatibility and support are offered by the OpenFlow [29], [50], protocol, which is widely acknowledged in the Software-Defined Networking (SDN) space. But due of its inflexible construction, it has little adaptability. It also raises certain scalability and security issues. The more contemporary and flexible protocol P4 [13] allows for packet processing customization. It is not widely compatible and is more complicated. Apart from these, structured data models and transactions are provided by additional protocols like YANG and Net Conf, which facilitate network configuration management. They may not, however, have the wide adoption of OpenFlow or the substantial programmability of P4. Every protocol serves a different purpose for the network; OpenFlow is more standardized, P4 offers more control, while Net Conf or YANG focuses especially on configuration management.

Depending on the specific needs of the network, OpenFlow or P4 should be used for fog offloading. Because of its compatibility and standardization, OpenFlow is a good fit for environments where stability and interoperability are important. On the other side, circumstances requiring specialized and tailored data handling benefit from P4's ability to specify packet processing actions. Therefore, P4 would be a better option if the job of offloading fog demands highly specialized packet processing and flexibility. However, OpenFlow could be the better choice for more extensive and standardized implementations. The optimal SDN protocol for fog offloading will depend on the specific requirements and characteristics of the fog computing environment. Although Netconf is well recognized for its user-friendliness and simplicity, which makes it a good fit for straightforward network management, OpenFlow is well known for its flexible and fine-grained control, which makes it great for complex and dynamic network administration. A network's complexity, scalability, necessary functionality, and knowledge base are some of the factors that influence the selection of the optimal SDN protocol. Other protocols have benefits and drawbacks, such as ONOS, BGP-SDN, and Faucet. Certain fog computing architectures may even mix these protocols to handle various parts of network management and job offloading, depending on their specific needs.

4) SDN WITH MACHINE LEARNING

For fog offloading, the combination of SDN and machine learning may provide a number of benefits. SDN provides a centralized view of the network, making it possible to optimize the process of choosing the best Fog node for offloading. Compute offloading techniques that improve Fog computing's reliability and efficiency may be developed using machine learning. Improved performance metrics such as latency, energy consumption, and quality of service, as well as the capacity to dynamically allocate services based on goals like power consumption, security, and QoS constraints are among the benefits of combining Software-Defined Networking (SDN) with Machine Learning (ML) for Fog offloading. Still, it is important to consider the disadvantages as well. The intricacy of the system might lead to higher maintenance costs. Obtaining a large amount of data to train machine learning models might be a major challenge. There may be a risk to the integrity and confidentiality of the data that the fog nodes process. Fog offloading may benefit from the use of lightweight techniques like federated learning to overcome these obstacles.

C. *METRICS FOR VALIDATION*

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Route selection, delay, and energy metrics are important during fog offloading; nevertheless, Table 2 indicates that not all research papers include these components. Research emphasis, resource constraints, and issue complexity are some of the factors that lead to this omission. Some research [50], [73], [83], and [98] focus just on delay optimization, ignoring energy issues or routes, whereas the majority only consider delay optimization and ignore metrics like latency or route selection. Path selection has been the focus of many studies, including [2], [69], [74], [85], and [97]. Sometimes, just a subset of these measures are explored by researchers due to study scope or experimental limitations. But ignoring any of these steps may have a big effect on how well the system works. IoT and Fog device batteries may run out if energy economy is neglected, while system performance and user experience may suffer if latency is ignored.

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Neglecting route selection might result in increased network congestion and subpar data throughput. Therefore, while building an effective fog offloading system, taking into account these factors is still crucial for achieving optimum system performance. Time-related problems are only partly addressed in previous frameworks, and resources' computational power has to be more effectively used. An effective job management method is needed to handle this problem, since the computational strain increases the total delay time, deployment cost, and energy consumption of IoT devices or resource-enriched fog nodes. The AI approaches employed in these systems result in excessive latency and system overload, which hinder real-time processing.

D. SIMULATION LIMITATION

The majority of recently published publications have only used approaches [13], [14], [2] and [50] that have been tested and evaluated in simulated scenarios. In addition, a variety of simulator types have been used, each having advantages and disadvantages of its own. The complexity of various network topologies, especially SDN, and the variety of simulators available will pose a serious obstacle to the practicality of the recommended techniques. Real-world validation is crucial even when simulations run using simulators like i-Fog Sim, Mininet, and Mat-lab provide informative findings. When applying simulation to real-world scenarios, there may be unforeseen challenges or novel considerations. Despite the abundance of theoretical concepts, there may not be as much room for the real-world evaluation or practical implementation of these strategies. Verifying the effectiveness and functioning of suggested algorithms requires extensive testing under a range of scenarios and real-world deployment.

E. DATASET

The need for consistency in the datasets used to predict and carry out fog-offloading research and publishing is one of the main challenges. This makes comparing the results of different experiments and generalizing the data difficult [101]. For the simulation's results to be supported, more actual data are required. This might lead to inflated hopes and inaccurate projections [101]. The specific use case and the surrounding environment also have a big influence on how well fog computing systems work. Because of this, it's crucial to choose the simulation tools and datasets that are appropriate for the given use case carefully [102]. Most studies either fail to identify the application as data and tasks that are assigned to other fog nodes, or they utilize randomized data. However, only a small number of studies, like [14], have made use of actual datasets. As such, most experiment results may not be relevant to real-world applications that need real-time execution or minimal latency. One major obstacle is the lack of relevant information and a modelling environment that appropriately captures the complex structure of the fog space.

F. SCALABILITY AND THROUGHPUT

The effectiveness of the system decreases noticeably as the number of nodes in the network rises. One major challenge facing the system will be the management of large networks of fog. There can only be sixteen fogs in the method mentioned in reference[2], and there is only one SDN controller in operation. Only five fogs were employed in [29], [90], and there were only two fogs per actual testbed node in [50], and six fogs in [13]. Although additional fogs have also been provided for other methods, only a small number of fogs have been tested in constrained areas and networks. It has been shown that efficiency decreases significantly as the number of fogs rises. The total amount of time spent waiting for jobs was decreased in the studies by using additional fog nodes [13]. One of the primary metrics of relevant study that stands in front of system efficiency is delay or waiting time, thus is important to take this into account. Scalability is a major problem, especially in systems with many of devices and fogs. The practicality of high overhead techniques in real-world applications may be impacted by the requirement for assistance in scaling effectively. This may apply to resource allocation, computation overhead, or communication overhead. A comprehensive evaluation of the scalability of the proposed methods is necessary, especially in real-world smart city implementations where a substantially larger number of devices and fog resources may be present.

G. SECURITY AND PRIVACY

Ensuring data security and privacy throughout offloading processes is a crucial yet challenging aspect. Strong security precautions may be disregarded by many contemporary methods, endangering sensitive data. The [12] integrates blockchain, SDN, and fog computing to provide an IoT security framework for agricultural. SDN controllers and blockchain technologies are used in this architecture to priorities secure IoT connection. This improves reliability

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and security especially for low-resource devices. Users' private information might be stolen by malicious users if smart healthcare systems' fog nodes don't have strong security measures in place. Furthermore, fog computing has to handle new problems like IoT devices with restricted resources and insider attacks. To mitigate these challenges, [7] proposes the use of SDN-based intelligent healthcare to provide a secure fog node authentication mechanism. To confirm the legitimacy of the fog node, the system implements an authentication technique in the SDN gateway. To lessen their processing load, IoT devices simply need to provide their functional characteristics and privacy to the SDN gateway. Secure communication, authentication, and permission must come first in fog offloading scenarios since the network environment is dynamic. However, these factors have not been considered and

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have been disregarded in most earlier studies. Through its centralized control structure, Software-Defined Networking (SDN) facilitates more efficient administration of these problems. The centralized approach makes it possible to apply security policies and network configurations in a dynamic, adaptable way that can adapt to changing network conditions. conditions and dangers. Improved network visibility is made possible by SDN and is essential for identifying and resolving abnormalities and security breaches. SDN unifies control, making it easier to integrate intrusion detection systems, firewalls, and encryption. On the other hand, decentralized techniques are required to guarantee security in fog offloading circumstances where SDN is not present. This entails setting up intrusion detection systems and distributed firewalls across several network nodes. Resilient encryption algorithms and efficient peer-to-peer authentication strategies are critical components of successful communication in these kinds of scenarios. Security rules must be manually adjusted and updated on a regular basis to keep up with the ever-changing network environment. This method is less flexible in quickly changing conditions than SDN-enabled systems and requires more coordination across different network components. On the other hand, it makes decentralized security possible, which might be useful in circumstances when centralized control is not feasible. The evolving fog offloading situation necessitates a comprehensive security architecture that incorporates secure communication, dynamic authentication, robust authorization, network segmentation, monitoring, blockchain integration, regular upgrades, and user education. Each component is essential to maintaining the fog environment and adapting to its ever-changing conditions.

H. OTHER CHALENGES

Table 2 shows that cloud offloading is not included in the majority of recent research' simulations and implementations. In many cases, this omission might cause delays since there isn't a thorough and precise integration of cloud services. The effectiveness of such algorithms in assessing cloud offloading is left unresolved if the cost and latency connected with cloud offloading are lower than those of using other fog nodes. As a consequence, jobs that are merely offloaded to other fog nodes rather than the cloud may incur higher costs and delays, which may decrease system efficiency. The fog environment is dynamic and made up of several fogs with different processing and memory capacity. On the other hand, the current jobs differ in terms of magnitude and due dates. Most systems and algorithms confront the difficulty of intelligently organizing these jobs to maximize the capacity and efficiency employed by the fog computing infrastructure. To illustrate, given three different fogs with different capacities and three task sizes (small, medium, and large), the best course of action would be to assign larger tasks to the fog with the highest capacity, medium tasks to the fog with the appropriate capacity, and smaller tasks to the fog with the sufficient capacity. Let's say a fog with higher processing power gets assigned to a little job for any reason, such as when it first appears. In such scenario, other fogs' computing capability would not be enough for the massive work, which might cause a delay in task completion since there wouldn't be a suitable fog with enough capacity. Although these factors are included in several resource and task management algorithms, concurrent resource and task management to maximize task offloading efficiency and load balancing remains a significant and inevitable challenge in large- scale systems and real-time tasks that are rarely addressed in research. Using machine learning methods in conjunction with SDN is one possible solution to improve fog systems load balancing and overcome this challenge. SDN may get up-to-date information about upcoming assignments and fog resources. Then, machine learning may choose the best fog resource for offloading by using the information stored in SDN to predict upcoming activities and events. The job size and fog capacity were considered to be the same in the majority of earlier investigations. Nevertheless, they have used randomly produced data in place of actual tasks, which is unable to adequately capture the current status of the fog network with regard to task transmission, task processing, and real-time reaction. Filling up these gaps emphasizes the need for more adaptable and thorough offloading strategies in the field of fog computing. Energy efficiency, scalability, security, adaptability, and latency in dynamic situations must all be balanced in fog offloading systems to take into account real-world implementation scenarios.

VI DISCUSSION

Tasks that an overloaded fog is unable to perform should be moved to another fog (a process called fog-to-fog offloading). Finding and selecting the ideal destination node with sufficient processing capability in the shortest amount of time is essential for real-time computing. Moreover, the majority of models ignore the variability seen in computer infrastructure. When a fog becomes overloaded, the majority of current algorithms request that the central server introduces the destination node for offloading. The central server assesses and contrasts all of the available nodes

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before choosing which one to install. Requesting the central server, running the decision process, replying to the first node, and finally uploading data to the target node all take a lengthy time.

Additionally, it is challenging to determine the best way to transfer data from the main fog to the target fog since the chosen route could be delayed by heavy traffic. Another fog node with an appropriate and adequate bandwidth for data transmission may be utilized if traffic congestion occurs o n a certain route. Making a decision about selecting fog with sufficient resources in the least amount of time and selecting an appropriate data transmission method are among the challenges in fog-to-fog offloading.

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Older frameworks examine time-related problems from a narrow angle, and they may make better use of computational power. Since computational strain increases the total delay time, deployment costs, and energy consumption of IoT devices or resource-enriched fog nodes, implementing an effective job management technique is necessary to overcome this problem.

The AI algorithms used in these methods cause a great deal of latency and overload into the system, which hinders real-time processing. Many aspects of fog-to-fog algorithms using SDN should be made better. One of the biggest challenges in the decision-making process and destination is fog selection, which necessitates a time-consuming and expensive decision-making algorithm for real-time processing. However, a fundamental challenge for most of these algorithms is that increased fog and network traffic leads to a decrease in system efficiency. It is thought that the network data plan fails to choose the connection for offloading to the intended location. When selecting a fog, connection traffic and data plans should also be carefully considered, since this might result in a delay in data transfer to the target fog. The offloading procedure will stop and fail if the connection breaks. TABLE 4 provides an overview of the techniques used in task offloading strategies. Concerns and challenges related to fog-to-fog offloading need research into and improvement of decision selection and route selection algorithms with suitable utilization of SDN-based networks' capabilities. It efficiently uses SDN network features and capabilities to speed up and improve the fog offloading algorithms, and artificial intelligence plays a key role in this process. Many of the many fog offloading strategies that have been seen have shifted to using AI as a result of the development of artificial intelligence and machine learning across numerous industries. Although these methods have been quite effective in increasing estimates for fog and clouds, they also come with a number of challenges because of their high computational cost and the restricted computing resources available in the network for fog. Obstacles such as:

1. Although fog reduces processing resources, they are sufficient for calculating algorithms related to artificial intelligence, deep learning, and machine learning.

2. In contrast, fog offloading is more difficult due to the bandwidth and communication load needed to transfer data between devices for offloading choices. A method for optimizing traffic dumping that is based on connection prediction is referred to as "link prediction-based." The link prediction-based approach prioritizes traffic unloading while maintaining quality of service (QoS) by choosing relevant seed nodes for effective data delivery. Finding a balance between reducing time delays and traffic dumping is essential for maximizing overall performance [103]. The best connection and node to offload may be found with the use of link prediction techniques.

One crucial aspect that remains constant throughout many fogs is the heterogeneity or homogeneity of fog nodes, a topic that might be explored further in future resource management studies. Partial or complete job offloading is another topic for future research on task management. Depending on the quantity and importance of the jobs, the algorithms should split and distribute them effectively. Each job may offload entirely or partly, depending on the fog processing limitations and task size.

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Author	Offloading type			Architecture			Protocol	Objective			Algorithm Type	Prediction		Implementation	
	IoT to Fog	Fog to Fog	Fog to Cloud	Centralized (No SDN)	Distributed	SDN		Path selection	Fog Selection	Energy		Fog Selection	Path	Real Test bed	Simulation
[2]		✓				✓	OpenFlow	✓	✓		Mathematical (Exhaustive search)				✓
[14]	✓	✓			✓	-	-		✓		Machin Learning (Neural Network)			✓	
[20]	✓					✓	OpenFlow		✓		Mathematical (Heuristic)				✓
[37]	✓		✓		✓	-	-		✓		Mathematical (Optimization)				✓
[82]	✓	✓			✓	-	-		✓		Mathematical (Matching Theory)				✓
[95]		✓				✓	-		✓		Mathematical (optimization problem)				✓
[13]	✓					✓	P4		✓		Mathematical (optimization)			✓	
[50]	✓					✓	OpenFlow			✓	Mathematical (Heuristic)			✓	
[29]		✓				✓	OpenFlow		✓		Machin Learning (Q-Learning mixed greedy)	✓		✓	
[90]	✓					✓	OpenFlow		✓		Mathematical (Greedy-Heuristic)				✓
[31]	✓					✓	Not specified				Mathematical (Optimization)				✓
[68]	✓					✓	OpenFlow		✓		Mathematical(heuristic)				✓
[11]	✓					✓	OpenFlow		✓		Mathematical(optimization)				✓
[70]	✓					✓	Not specified		✓		Q-Learning				✓
[91]	✓					✓	OpenFlow		✓		Mathematical			✓	
[69]	✓					✓	OpenFlow	✓			Machine learning				✓
[92]	✓					✓	Not specified		✓		Q-Learning				✓
[93]	✓					✓	Not specified		✓		Mathematical				✓
[94]	✓					✓	Not specified		✓		Greedy (Heuristic)				✓
[30]		✓			✓	-	-		✓		Blockchain				✓
[96]	✓	✓		✓		-	-		✓		Mathematical				✓
[22]		✓			✓	-	-				Mathematical				✓
[83]	✓				✓	-	-		✓	✓	Mathematical (Weighted Potential Game)				✓
[23]		✓	✓		✓	-	-				Mathematical	✓			✓
[36]	✓	✓	✓		✓	-	-		✓		Mathematical				✓
[89]		✓			✓	-	-				Mathematical				✓
[85]	✓	-	-		✓	-	-	✓	✓		Mathematical			✓	
[97]	✓			✓		-	-	✓	✓	✓	Q-Learning		✓		✓
[98]	✓		✓	✓		-	-		✓	✓	Mathematical				✓
[73]		✓				✓	Not specified		✓	✓	Differential Evolutionary				✓
[74]		✓				✓	OpenFlow	✓	✓		Heuristic greedy algorithm				✓

Future research should take into account energy consumption and algorithm computing cost, which are crucial parameters for fog offloading strategies but are often disregarded in many recommended algorithms, in addition to time and delay.

SDN offers a lot of advantages, but it also has certain disadvantages. Since distant fog nodes or data center's always add delays, latency is still a problem. Adoption may be hampered by the complexity of SDN implementation, potential security risks from centralized administration, and higher network overhead. Some organizations may be discouraged from utilizing SDN for fog offloading due to the initial cost of specialized hardware and software, and

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interoperability issues may arise, particularly in heterogeneous fog computing environments. The suitability of SDN must be determined by closely analyzing each use case and set of r equirements. Some of the disadvantages of fog computing simulation include model accuracy problems, scalability problems, the absence of real-time aspects, cost, abstraction, and reliance on behavioral assumptions.

Simulating the dynamic nature of fog computing successfully may be challenging due to the continuously shifting workloads and network conditions. Low predictive power, issues with validation and verification, and imperfect network models are further obstacles. Due to resource constraints, large-scale and resource-intensive

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simulations may also be challenging to run. While simulations may provide valuable insights, it is advisable to exercise care when using them and to interpret the results in light of these constraints. Combining simulation and real-world testing might result in a deeper understanding of fog computing systems.

VII CONCLUSION AND FUTURE WORK

The majority of relevant fog offloading algorithms that have been studied concentrate on one or two of these three crucial elements rather than managing resources, tasks, and task offloading holistically within the same system. Implementing these algorithms in complex real-world settings with a range of jobs and different fogs hardware and software with fluctuating resources at the same time will provide a number of challenges for Lake of this perspective. Fog offloading solutions thus need more research in order to be highly adaptive to the wide range and comprehensive real-time jobresponsiveness of IoT. SDN offers a number of benefits for computer system fog unloading. By allowing dynamic and customizable designs to adjust to shifting workloads, it facilitates network flexibility. Centralized control simplifies decision- making in order to improve service quality, optimize routing, and offload tasks. Because SDN enhances traffic optimization and grows to accommodate the growing number of IoT devices and fog nodes, it is an essential tool for real fog computing. SDN is able to efficiently manage the fog network. However, the likelihood of early implementation costs is increased by its centralized and distinctive design. If fog processing uses considerably less computational resources while transferring minimal data over the network, machine learning could be useful. The decision-making procedure for selecting the fog locations for offloading and the connecting link for data transfer may be determined by machine learning. The issue may be solved by choosing an effective machine learning technique that reduces the computational load on the network with its limited connections and processing resources. Thus, future research endeavors seek to optimize and expedite the fog-to-fog algorithm via the effective utilization of SDN networks' capabilities and artificial intelligence's traits. Moreover, it is feasible to deploy more SDN controllers and fog networks on a larger scale and to carry out thorough testing in actual networks and situations. Similarly, investigating how SDN networks may be integrated with centralized and decentralized networks outside of SDN may be a workable way to accept and combine new and larger networks as well as difficulties related to smart cities.

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CONFLICTS OF INTEREST

The authors declare no conflicts of interest regarding the publication of this paper.

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